Deep Learning-Based Waste Classification and Automated Segregation Using Convolutional Neural Networks

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***Abstract*—Mixed waste lowers recycling rates and increases the burden on landfills, and manual sorting is slow, costly and unsafe. This paper presents a deep learning-based approach to waste classification and segregation using convolutional neural networks (CNNs). The method uses transfer learning on a pre-trained CNN to classify waste images into six classes, and a confidence-based decision rule to assign each item to the correct segregation class or to a manual-check class when the prediction is uncertain. The dataset, pre-processing, augmentation, training strategy and evaluation metrics are described, and common CNN models are compared. The study shows that transfer learning with a lightweight CNN is a practical way to build an accurate and low-cost waste segregation system, provided that the training data reflect real waste conditions.**

***Index Terms*—Artificial intelligence, CNN, deep learning, image classification, transfer learning, waste classification, waste segregation.**

I. INTRODUCTION

Large quantities of solid waste are produced every day in cities, and only a small part is separated properly at source. When recyclable and organic waste are mixed, the recyclables become dirty and are often sent to landfill. Manual sorting is possible, but it is slow, depends on the skill of workers, and exposes them to sharp, wet and hazardous items.

Computer vision offers a better way. A camera image of a waste item can be analysed by a CNN, which learns visual features such as colour, texture, shape and transparency directly from examples. If the item is identified correctly, it can be sent to the right bin automatically. Deep learning has therefore become the main technique for waste classification research [1], [6].

The contributions of this paper are: (a) a review of CNN models used for waste classification; (b) a transfer-learning methodology for a six-class waste classifier; (c) a confidence-based segregation rule that reduces wrong sorting; and (d) an evaluation plan with suitable metrics.

II. RELATED WORK

Datasets. The TrashNet dataset of Yang and Thung [1] has about 2,500 images in six classes: glass, paper, cardboard, plastic, metal and trash. It is a popular benchmark, but its images show one item on a plain background. Majchrowska et al. [6] studied waste detection in natural and urban scenes and showed that cluttered backgrounds, dirt and deformed items make the task much harder.

Transfer learning and augmentation. Pan and Yang [7] describe transfer learning as reusing knowledge learned on a large source task for a smaller target task, which suits waste classification where labelled data are limited. Shorten and Khoshgoftaar [8] survey image augmentation methods that reduce over-fitting by creating varied copies of training images.

CNN models. ResNet [2] introduced residual connections that make very deep networks easier to train. MobileNet [3] uses depthwise separable convolutions to reduce computation, and EfficientNet [4] scales depth, width and resolution together. For images with several items, detectors such as YOLO [5] predict boxes and classes in one pass. Table I compares these models.

TABLE I
CNN MODELS USED FOR WASTE ANALYSIS

Model	Key idea	Suitability
ResNet [2]	Residual connections allow very deep networks	High accuracy; heavier to run
MobileNet [3]	Depthwise separable convolutions	Light; good for small devices
EfficientNet [4]	Balanced depth, width, resolution scaling	Good accuracy per parameter
YOLO [5]	Single-pass detection with boxes	Several items per image

Table II places this work beside two key earlier studies.

TABLE II
COMPARISON WITH EARLIER STUDIES

Study	Data / Scene	Main Focus
Yang and Thung [1]	Single items, plain background	Six-class benchmark dataset
Majchrowska et al. [6]	Natural and urban scenes	Waste detection in clutter
This work	Public + local images	Transfer learning with confidence-based segregation and manual-check class

III. PROPOSED METHODOLOGY

The overall flow is shown in Fig. 1. An input image is pre-processed, passed through a CNN and converted to class probabilities by a softmax layer. A decision rule then selects the segregation output.

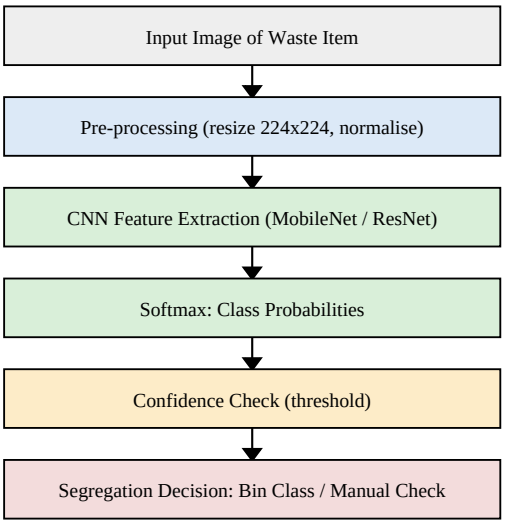

Fig. 1. Classification and segregation flow of the proposed method.

A. Dataset and Classes

Images from TrashNet [1] are combined with locally collected photographs of items common in the target area, such as food waste, plastic pouches and leaves. Images are grouped into the six segregation classes of Table III. Local images are valuable because they bring in background, lighting and item variety that public data lack.

TABLE III
WASTE CLASSES AND SEGREGATION OUTPUT

Class	Examples	Segregation Output
Organic	Food scraps, leaves	Compost bin
Paper / Cardboard	Newspaper, boxes	Paper recycling bin
Plastic	Bottles, wrappers	Plastic recycling bin
Metal	Cans, foil	Metal recycling bin
Glass	Bottles, jars	Glass recycling bin
Other / Uncertain	Mixed or unclear items	Manual-check bin

Data collection guidelines. Photograph each item from several angles and distances, on different backgrounds and in different light. Include clean, dirty, wet, crushed and partly hidden items. Remove duplicate and blurred images, and have a second person check labels, because wrong labels directly reduce accuracy. Keep the number of images per class as balanced as possible.

B. Pre-processing and Augmentation

Each image is resized to 224×224 pixels and its pixel values are normalised. During training, random rotation, horizontal flip, zoom and brightness change are applied. Augmentation increases the effective size of the dataset and helps the model cope with different camera angles and light [8].

C. Model and Training

Following the transfer learning idea [7], a pre-trained CNN, for example MobileNet [3], is used as a feature extractor. Its original output layer is replaced with a new layer for the six classes. In the first stage only the new layers are trained; in the second stage some deeper layers are unfrozen and trained with a small learning rate. The dataset is split into training, validation and test sets, and early stopping is used to avoid over-fitting. The loss function is categorical cross-entropy,

$$L = -\Sigma\, y_i \log(p_i), \qquad (1)$$

where y_i is the true label and p_i the predicted probability of class i.

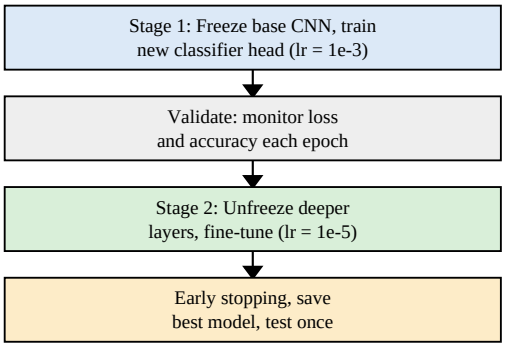

Fig. 2. Two-stage training workflow.

Table IV lists suggested settings. These values are a starting point and should be tuned using the validation set.

TABLE IV
SUGGESTED TRAINING SETTINGS

Parameter	Suggested Value
Input size	224 × 224 × 3
Base model	MobileNet (ImageNet pre-trained)
Optimizer	Adam [9]
Learning rate	1e-3 (stage 1), 1e-5 (stage 2)
Batch size	32
Epochs	20 (stage 1) + 10 (stage 2)
Dropout	0.3 before final layer
Loss function	Categorical cross-entropy
Confidence threshold τ	0.80

D. Confidence-Based Segregation Rule

Algorithm 1 gives the steps. Let $p_{\max}$ be the highest softmax probability. If $p_{\max} \geq \tau$ (for example $\tau = 0.80$), the item is assigned to the predicted class. Otherwise it goes to the manual-check class and its image is saved for later labelling and retraining. This rule trades a small number of manual checks for a lower rate of wrong sorting.

Algorithm 1. Confidence-based segregation

Input: image x, trained CNN f, threshold τ
1. $x' \leftarrow$ resize and normalise(x)
2. $p \leftarrow$ softmax($f(x')$)
3. $c \leftarrow$ argmax(p); $p_{\max} \leftarrow$ max(p)
4. **if** $p_{\max} \geq \tau$ **then** output class c
5. **else** output "Manual check" and save x
Output: segregation class

Illustrative example. Suppose the softmax output for an image is plastic 0.91, glass 0.05, metal 0.02 and others 0.02. Since 0.91 ≥ 0.80, the item is sent to the plastic recycling bin. For another image the output is paper 0.48, cardboard 0.43 and others 0.09. Here the highest value is below 0.80, so the item goes to manual check and its image is stored for labelling. These values are only an example to explain the rule.

IV. EVALUATION PLAN

Performance on the test set is measured by accuracy, precision, recall and F1-score:

$$\text{Accuracy} = (TP+TN)/(TP+TN+FP+FN),$$
$$F1 = 2\cdot\text{Precision}\cdot\text{Recall}/(\text{Precision}+\text{Recall}). \qquad (2)$$

A confusion matrix shows which classes are mixed up, such as paper and cardboard or glass and plastic. Additional measures are inference time per image, the share of items sent to manual check, and the wrong-sorting rate at the chosen threshold τ. Testing should also include images with dirt, wetness and overlapping items to match real conditions [6].

V. EXPERIMENTAL SETUP

The experiments proposed for this work are listed in Table V. The dataset is divided into 70% training, 15% validation and 15% test images, with the split kept the same for all models so that comparison is fair. Training can be done on a laptop with a GPU or on a free cloud notebook, using Python with TensorFlow or PyTorch and OpenCV for image handling. Each experiment is repeated with different random seeds, and the mean and spread of the metrics are reported.

TABLE V
PROPOSED EXPERIMENTS

Experiment	What is compared
E1: Baseline	Small CNN trained from scratch
E2: Transfer learning	MobileNet / ResNet / EfficientNet fine-tuned
E3: Augmentation	With and without augmentation [8]
E4: Threshold τ	τ = 0.6, 0.7, 0.8, 0.9 vs. wrong-sorting and manual-check rate
E5: Local data	Public data only vs. public + local images

Experiment E1 gives a baseline, so that the gain from transfer learning in E2 is clear. E3 and E5 show how much augmentation and local images help. E4 studies the effect of the threshold τ: a high value gives fewer wrong sortings but sends more items to manual check, so the best value depends on how costly each type of error is.

VI. APPLICATIONS AND IMPACT

Households and housing societies: a classifier on a phone or small computer can guide residents to place items in the right bin. **Schools and campuses:** the system can support awareness programmes and make segregation a daily habit. **Recycling and sorting centres:** an image classifier can assist workers on sorting lines and reduce their contact with hazardous items. **Municipal planning:** counts of items by class can show which kinds of waste dominate an area and help plan recycling programmes.

Environmental benefit. Cleaner segregation raises the quality of recycled material, lowers the volume sent to landfill and reduces the need for new raw material. **Social aspects.** Automation should support waste workers and not replace their livelihood without alternatives; training them to operate and maintain such systems is a good way forward. **Ethics and privacy.** Images should capture only waste items, not people, and should be kept only as long as needed for model improvement.

VII. ERROR ANALYSIS AND MITIGATION

Even a good classifier makes mistakes, and knowing their causes helps to reduce them. Table VI lists errors that are common in waste classification and ways to reduce them. After testing, the confusion matrix and a sample of wrongly predicted images should be studied by hand, so that new training images can be collected for the weak cases.

TABLE VI
COMMON ERRORS AND MITIGATION

Common Error	Likely Cause	Mitigation
Paper vs. cardboard	Similar colour and texture	More close-up images; higher resolution
Glass vs. clear plastic	Transparency, reflection	Varied lighting data; manual-check class
Wet or dirty items	Not seen in clean datasets	Add local dirty and wet samples
Crushed or folded items	Shape change	Rotation, zoom and shear augmentation
Rare classes missed	Class imbalance	Class weights; oversampling

Threats to validity. Results on public data may not hold on a different city's waste. Test images that are very similar to training images can give a falsely high accuracy, so the test set should be kept separate and should contain images from different sessions and places. Reporting per-class metrics, and not only overall accuracy, avoids hiding poor results on rare classes.

VIII. DISCUSSION

Advantages. Transfer learning needs far less data and training time than building a CNN from scratch. Lightweight models such as MobileNet can run on low-cost computers. Automatic sorting reduces human contact with waste and improves the purity of recyclable material.

Limitations. Models trained on clean, single-item images often lose accuracy on real waste [6]. Visually similar items, such as clear plastic and glass, are hard to separate using images alone. Rare classes have few examples, which can bias the model toward common classes. Addressing these needs larger local datasets, class-balanced training and regular retraining using the manual-check images.

IX. CONCLUSION AND FUTURE SCOPE

This paper presented a CNN-based method for waste classification and segregation using transfer learning and a confidence-based decision rule. The approach is simple, low-cost and suitable for small devices, and the manual-check class makes the system safer against wrong sorting. Future work includes: (i) building a large local dataset with cluttered, wet and dirty items; (ii) using object detection to handle several items in one image; (iii) applying model compression and quantisation for faster inference on low-power computers; (iv) adding more classes such as e-waste and hazardous waste; (v) using self-training so that manual-check images improve the model over time; and (vi) testing the method on real waste streams and reporting accuracy, speed and cost together.

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